Hackathon Problem Statements

Regd No: 25091A3279, 25091A32K3

Problem Statement No: 22

Medicine Expiry & Stock Management System Design a pharmacy inventory system that uses a min-heap to always surface medicines nearest to expiry for priority dispensing (FEFO — First-Expiry-First-Out), reducing wastage. Apply demand forecasting using time-series models to predict restocking needs for essential medicines. Represent medicines, batches, and suppliers using OOP class hierarchies. This is especially valuable for government dispensaries that frequently face both stockouts and expired medicine wastage simultaneously.

Regd No: 25091A32P0, 25091A3272

Problem Statement No: 44

Building Fire Evacuation Route Planner Build an evacuation route planning tool using BFS/DFS on a graph representation of a building's floor plan to compute the fastest evacuation paths from any room to the nearest exit, accounting for blocked routes during simulated fire scenarios. Use OOP to model rooms, exits, and hazard zones as graph nodes with dynamic states. This tool can be integrated into building safety systems for schools, hospitals, and public buildings.

Regd No: 25091A32A9, 25091A32A8

Problem Statement No: 50

Smart Bin Fill-Level Monitoring & Dispatch System Build a waste bin monitoring system using a min-heap to prioritize collection trucks toward the fullest bins first, based on simulated fill-level sensor data across a city. Combine with route optimization to minimize travel distance for collection rounds. Use OOP to model bins, trucks, and collection zones. This prevents overflowing bins and reduces unnecessary trips to partially-full bins, improving sanitation and operational efficiency.

Regd No: 25091A32J7, 25091A32M4

Problem Statement No: 60

Student Plagiarism Detection Tool Build a plagiarism detection tool using string-matching algorithms (KMP or Rabin-Karp) to compare submitted assignments against a reference corpus and flag suspiciously similar sections. Compute a similarity score and highlight matched text spans for instructor review. Use OOP to model documents, comparison reports, and similarity thresholds. This helps maintain academic integrity in an era of easy content copying.

Regd No: 25091A32F1, 25091A32F0

Problem Statement No: 150

Encrypted Community Bulletin Board System Design a secure community bulletin board system using cryptographic hashing for message integrity verification combined with an OOP design modeling users, posts, and moderation roles. Implement basic encryption for sensitive community postings (like neighborhood watch alerts). This demonstrates how secure, trustworthy digital community infrastructure can be built even with foundational-level cryptography concepts.